Layered Architecture

Data Network

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1 ?

1.1 Nuts & Bolts

host(end system) **communication link** **packet switch** .

- **Host**: PC, ,
- **Communication link**: fiber, copper, radio, satellite — **bandwidth(, bps)**
- **Packet switch**: — **router**(network core), **switch**(link layer)

"network of networks" , ISP(Internet Service Provider) .

1.2 Service

(API) .
Web, VoIP, email, , e-commerce .

1.3 Protocol

Protocol: , ,

- : TCP, IP, HTTP, 802.11 (WiFi)
- **RFC(Request for Comments)** **IETF(Internet Engineering Task Force)**

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2 Network Edge & Access Networks

2.1 3

Network edge	hosts(clients & servers),
Access network	end system edge router

Network core	mesh

2.2 Host

Host **L bits** .

$$d_{trans} = \frac{L \text{ (bits)}}{R \text{ (bits/sec)}}$$

R link transmission rate(= link capacity = link bandwidth) .

2.3

- *DSL/* : , HFC(Hybrid Fiber Coax)
- *Ethernet*: / , 10 Mbps ~ 10 Gbps

- *Wireless LAN (802.11 WiFi)*: ~100 ft, 11/54/450 Mbps
- *Wide-area wireless*: tens of km, 1~10 Mbps (3G, 4G LTE)

2.4 (Physical Media)

Twisted pair (TP)	Cat5: 100 Mbps/1 Gbps, Cat6: 10 Gbps
Coaxial cable	, (broadband), HFC
Fiber optic	, 10s~100s Gbps,
Radio	, / /

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3 Network Core

3.1 Packet Switching

store-and-forward .

3.1.1 Store-and-Forward

.

$$d_{end\text{-}end} = N \cdot \frac{L}{R} \quad \text{(N hop, propagation } \quad \text{)}$$

3.1.2 Queueing Delay & Loss

- $>$ $\rightarrow$
- **drop( )**

3.1.3 Two Key Functions

Forwarding	$\rightarrow$	(local)
Routing	-	(, global)

3.2 Circuit Switching

/ end-to-end **(dedicated)** .

3.2.1 FDM vs TDM

FDM (Frequency Division Multiplexing)	
TDM (Time Division Multiplexing)	

3.3 Packet vs Circuit Switching

	Packet Switching	Circuit Switching
	(on-demand)	(dedicated)
		(: 10 /1 Mbps)
QoS		
	Bursty data	/

3.3.1 —

1 Mb/s , 1 100 kb/s, 10%

- Circuit switching: **10**
- Packet switching: **35** , P(10) < 0.0004

$$P(X > 10) = \sum_{k=11}^{35} \binom{35}{k} (0.1)^k (0.9)^{35-k} \approx 0.0004$$

$X \sim \text{Binomial}(35,\ 0.1)$, $E[X] = 3.5$

3.4 Internet Structure: Network of Networks

ISP $O(N^2)$.

$$\binom{N}{2} = \frac{N(N-1)}{2} \approx \frac{(10^6)^2}{2} = 5 \times 10^{11}$$

:

```
Tier-1 ISP   IXP   Tier-1 ISP
     |                  |
Regional ISP        Regional ISP
     |                  |
Access ISP          Access ISP
```

- **IXP (Internet Exchange Point)**: ISP peering
- **Content provider network** (Google, Akamai): Tier-1

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4 Delay, Loss, Throughput

4.1 4

$$d_{nodal} = d_{proc} + d_{queue} + d_{trans} + d_{prop}$$

d_{proc}	—	+	< msec
d_{queue}	$\propto \frac{\rho}{1-\rho}$		,
d_{trans}	L/R		.
d_{prop}	d/s		$(s \approx 2 \times 10^8$ m/s$)$

! d_{trans} vs d_{prop}

- d_{trans}: — 10 Gbps
- d_{prop}: — →

4.2 Traffic Intensity

$$\rho = \frac{La}{R}$$

- L: (bits)
- a: (packets/sec)
- R: (bits/sec)
- La:

ρ	
$\rho \approx 0$	, 0
$\rho \to 1$	, $\to \infty$
$\rho > 1$	→ **packet drop**

M/M/1 :

$$E[W] = \frac{\rho}{\mu(1-\rho)}, \quad \mu = R/L$$

4.3 Throughput ( )

$$\text{throughput} = \min(R_c,\ R_s,\ R/N)$$

R_c	
R_s	
R/N	N R per-connection

(bottleneck link) .

5 Protocol Layers & Service Models

5.1

.

1. —
2. — (transparent)
3. : ,

5.2 Internet 5-Layer Stack

```
Application    HTTP, FTP, SMTP, DNS

Transport      TCP, UDP

Network        IP,

Link           Ethernet, 802.11 (WiFi), PPP

Physical       bits "on the wire"
```

5.3 OSI 7

Internet Stack	OSI	
Application	Application	
( )	Presentation	, ,
( )	Session	,
Transport	Transport	Process-to-Process
Network	Network	Host-to-Host
Link	Data Link	Hop-to-Hop
Physical	Physical	

Internet Session/Presentation

End-to-End Argument (Saltzer, Reed, Clark, 1984): , end system . Presentation (,) Session ( ) Application .

	OSI	Internet
	Presentation	TLS/SSL
	Presentation	HTTP `Content-Encoding: gzip`

Session	HTTP Cookie, WebSocket

5.4 Encapsulation ( )

, .

$$M \xrightarrow{\text{Transport}} H_t|M \xrightarrow{\text{Network}} H_n|H_t|M \xrightarrow{\text{Link}} H_l|H_n|H_t|M|\text{FCS}$$

PDU		
Message	Application	—
Segment (TCP) / Datagram (UDP)	Transport	src/dst port, seq#, checksum
Datagram (IP Packet)	Network	src/dst IP, TTL, protocol
Frame	Link	src/dst MAC, type, FCS
Bit	Physical	—

5.5 Hop-by-Hop

L1~L3 , L1~L2 .

```
[ ]
  Physical: bits → Frame
  Link:     FCS   → H_l   (decapsulation)

[ ]
  Network:  H_n  dst IP → Forwarding Table
            TTL   (0   )

[  ]
  Link:       H_l'   (  hop  MAC   )
  Physical: bits
```

!

- H_l: hop (MAC)
- H_n: TTL , src/dst IP end-to-end (NAT)
- H_t, M**: — end-to-end

6 Network Security

6.1

“ ” . , .

6.2

6.2.1 Malware

Virus	,
Worm	( ) ,

botnet spam , DDoS .

6.2.2 DoS / DDoS

· bogus .

```
→ botnet →
```

6.2.3 Packet Sniffing

NIC **promiscuous mode** .

- —
- Broadcast medium **WiFi** ( )
- : (TLS, WPA3)

6.2.4 IP Spoofing

src IP . .

- DDoS
- : Ingress filtering (RFC 2827)

Sniffing	( )	(,)
Spoofing	( )	,

7

$$d_{nodal} = d_{proc} + d_{queue} + d_{trans} + d_{prop}$$

$$d_{trans} = \frac{L}{R}, \quad d_{prop} = \frac{d}{s}$$

$$\rho = \frac{La}{R} \quad \Rightarrow \quad E[W]_{M/M/1} = \frac{\rho}{\mu(1-\rho)}$$

$$\text{Throughput} = \min(R_c,\ R_s,\ R/N)$$

$$P(\text{packet switching} \quad) = \sum_{k=k_0}^{N} \binom{N}{k} p^k (1-p)^{N-k}$$

$$(\text{full mesh}) = \binom{N}{2} = \frac{N(N-1)}{2} = O(N^2)$$